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PLACE-LEARNING, COGNITIVE MAPS, AND PARSIMONY*

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A. INTRODUCTION

In recent years a number of experiments have been reported from Tolman's laboratories (8, 9, 12, 13, 14) dealing with the problem of "place" *vs.* "response" learning. These studies have had the effect of inducing a nostalgic atmosphere in that they hark back to a much earlier period in the history of psychology and reopen questions that have long since been assumed to have found adequate answers in classical animal experimentation. One need only recall the studies of Lashley and of his students and associates to realize that there exists a substantial body of evidence demonstrating that mammals in their maze and problem-box learning, do not develop specific "responses" with which to cope with the demands of the situation. Rather, in mastering a problem, they develop a wide range of approaches and orientations, so that upon being deprived of any given motor or sensory ability, they exhibit great versatility in the utilization of a variety of previously unused responses in arriving at a solution to the problem with which they are confronted.

As long ago as 1924, Lashley (5) showed that monkeys after the destruction of the precentral gyri mastered latch-boxes with the previously non-utilized arm. Shortly afterward, Lashley and McCarthy (6) demonstrated that rats subjected to cerebellar injuries after learning the maze habit literally "rolled" through the maze with great accuracy despite profound motor disorganization. Surely, this study reveals the lack of response specificity in maze learning. Still later, in his monograph *Brain Mechanisms and Intelligence*, Lashley reported some observations which so closely agree with the reports from Tolman's laboratories that they warrant quotation.

> With a number of normal animals which had learned the Maze III, I made the following test. The wire mesh cover was removed from the maze and starting compartment, and the entrance to the maze blocked. An animal was then placed in the starting-compartment and kept under observation. Of 20 so tested, 5 pursued the most direct course across the tops of the partitions to the food box, with practically no random exploration, 3 wandered about and reached the food seemingly by

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> chance, and the others dropped into the first alley and followed their customary path. The probability of such a result from chance cannot be computed, but the behavior of the 5 that followed the direct course to the food strongly suggested that they were perfectly oriented with respect to its direction, although they had never before reached it save by the indirect path of the maze. Observations of others upon the shortening of the path through the maze, and on a maze constructed to test this question of direct adaptation, likewise suggest that the rat forms, from the indirect path of the maze, some generalization as to the relative positions of the starting and food boxes" (5, p. 137).

Many other studies could be cited, but even from the few sources mentioned above it can be seen that the so-called "place" learning studies do not provide us with a *new* body of evidence, but rather reiterate in somewhat new situations a series of observations made more than two decades ago. It is legitimate, therefore, to ask what it is that is new in Tolman's recent work and to examine the validity of the new conceptual containers into which the old wine has been poured.

The motivation of Tolman and his associates in reopening the question of place learning is best expressed in the first papers of their series (12). In it they point out the need to distinguish Tolman's concept of *expectancy* from theories of a stimulus-response kind which assert that in a learning situation "what is learned is a tendency to make a particular response" (12, p. 14). They believe that "expectancy" means more than this; and, that what it implies is *not* the tendency to make a particular response after learning, but rather the existence in the animal of the expectation of food at a specific location of the goal box. Thus a rat which has learned a maze has not merely learned a series of responses, but has come to expect food at the location of the goal box. Therefore, if such an animal finds its customary path to the goal blocked, it will move to the location of the goal box along other paths if these are available. The delineation of the behaviors to be studied is expressed succinctly by Tolman, Ritchie, and Kalish in the following refined definition of the expectancy concept:

> If x is deprived of food and x has been trained on path P and x is now put on path P and path P is now blocked and there are other paths which lead away from path P, one of which points directly to location L, then (x runs down the path which points directly to location L—x expects food at location L) (12, p. 15).

They recognize that a definition does not need to be tested as to its truth or falsity, but rather that a search must be instigated to determine whether the class that has been defined has any members. They, therefore, pro-

ceeded to determine empirically whether or not there were "any rats which do in fact take the shortest path to the goal location, when the original path is blocked" (p. 15). If the finding of a "class member" had been their only objective it would, of course, have been possible for these investigators to have saved themselves trouble and time by referring to the data which Lashley (5) had already reported. However, it is clear from their discussion that the experiment had a broader general objective than the mere quest for an empiric. On the one hand the experiment was designed to call into question the adequacy of any *S-R* theory of learning, and on the other to advance the hypothesis that the rat develops certain "expectancies" which are the *essence* of what it learns. This latter construct has been since expanded by Tolman (10) into the theory that the rat develops a "cognitive map" of the environment and that its behavior indicates that the animal has "acquired not merely a strip-map to the effect that the original specifically trained-on path led to food but, rather, a wider comprehensive map to the effect that food was located in such and such a direction in the room" (p. 204).

The theoretical question involved in the "place" learning studies is therefore not a mere matter of definitions, but is rather the validity of the cognitive hypothesis as an explanatory principle. More specifically, in the Tolman, Ritchie, and Kalish study (12) behavior is observed which cannot be accounted for adequately by *S-R* theory. To account for the "place" rather than the "response" type of behavior Tolman (10) has advanced a theory of cognitive maps, in accordance with which the animal is presumed to develop more or less broad "cognitions" and "expectancies." Such an explanation involves the imputation of higher-level behavior on the part of the rat, and can only be accepted if explanations in terms of simpler psychological functions are found to be incapable of accounting for the behaviors that have been observed.

The cognitive theorists have repeatedly (8, 9, 15) used the principle of exclusion as the basis for their argument. They have assumed that once a behavior cannot be adequately accounted for in terms of an *S-R* theory, it can only be explained in cognitive terms. However, this method of residual analysis is inherently weak and can only be used as a tentative device when all possible available explanations are being considered. The correct rejection of an *S-R* explanation, however, does not exhaust the possibilities for offering simpler explanations. An additional explanation of the results of Tolman, Ritchie, and Kalish is suggested in the sensory-integration theory of learning which existed for some time (7) and has most recently been advanced by Birch and Bitterman (1, 2).

In terms of sensory-integration theory the animals of Tolman, Ritchie, and Kalish when running to the food always found the food to be in a fixed spatial and temporal relationship with a light. These conditions were therefore ideal for the development of a sensory-integration between the light stimulus and the food stimulus. If such an integration had occurred, then the light should have become the stimulus equivalent of food. Therefore, the animal, under given metabolic conditions (hunger) would rapidly come to run toward the light (or toward the lighter side of the apparatus) when its previously traversed path was blocked. Such an explanation would treat the "place" learning phenomenon as a simple instance of stimulus-equivalence formation and would involve no assumptions as to higher cognitive processes, "hypotheses" or "maps."

Of course all such conjecture is dependent upon whether the behavior that has been described by Tolman, Ritchie, and Kalish actually occurs. If, as Gentry, Brown, and Lee (4) and Gentry, Brown, and Kaplan (3) have claimed, the behavior does not occur, then there is nothing to explain. However, a sensory integration theory leads us to expect that such kinds of behavior should occur if a sufficient number of training trials are given the animal. For this reason the experiment to be reported here was conducted in two parts. Part I consisted of a series of tests of the light-food integration hypothesis after the animals had had the number of training trials described by previous experimenters. Part II consisted of the extension of the training trials well beyond the limits imposed by the previous workers, and the testing of the animals after such additional training.

B. Procedure

Throughout this study an attempt was made to duplicate the experimental features introduced by Tolman and his associates (12). The apparatus used is identical with that which is described by them, and the procedure of Experiment I is also identical except for the two light-test variations that are introduced. In Experiment II, the training program was extended beyond the limitations set in their study.

C. Animals

The females of six litters of animals used in this study, all Sherman strain albino rats of the City College colony, were from 7 to 8 months old at the start of the experimental program. Thirty-eight animals were employed initially, but two were discarded before training because of their very poor performance on the practice maze. Prior to the experimental

training situation the animals were run on a practice elevated maze for 14 consecutive days and were, therefore, maze-wise at the start.

D. Apparatus

The room in which the work was conducted was light-proof and completely screened with black cloth. Wood shavings were spread on the floor in order to provide a homogeneous background for the maze. The experimental training maze and the laboratory arrangement are schematically presented in Figure 1. The apparatus consisted of an unpainted circular table three feet in diameter, and unpainted pine pathways two inches wide. Path *AB*, the starting path, was 24 inches long, while Paths *CD*, *DE*, and *EF* were each 18 inches long. At the end of Pathway *FG*, which was 60 inches long, a wooden food-box, 10 x 4 x 6 inches, was fastened. A 7.5-watt bulb, in an ordinary goose-neck desk lamp was located 6 inches behind the food-box and the reflector of this lamp was turned in such a way that the light was primarily directed down Path *FG*. This was the only source of light in the room during the experimental training situation.

The apparatus which was used in the various test trials is shown in Figure 2. It consisted of the same starting pathway, circular tube, and the alley on Path *CD*. The food-box and the pathways leading away from the table (represented by the broken lines in Figure 2) were removed, and a block was placed inside Alley *CD*, 20 inches from the entrance on the table. Twelve 6-ft. and six 2-ft. unpainted pine pathways, all two inches wide, were placed around the table top in a semi-circular fashion starting from the perpendicular to the right of Path *AB*. The paths radiated around half the circumference of the table and were placed at angles of 10 degrees to each other. They were fastened in such a way that the table top could be rotated without any interference. The light which was located behind the food-box during training situation was left in place, and for the appropriate experimental tests an identical lamp was added in a corresponding position on the other side of the apparatus.

E. Experiment I

1. *Training of Animals*

Twenty-four hours before each of the series of trials was initiated, the rats were fed Purine Dog Chow pellets for one hour after which time all food was removed from the cage. On each of the four pre-test training days, the animals were given three practice trials. After each trial the table top was rotated 90 degrees. As can be seen, the training procedure is

identical with the one used by Tolman, and his associates (12). The daily procedure follows:

a. Day 1. On the first trial each animal was placed in the food-box for five minutes. On the second trial they were allowed to run into the food-

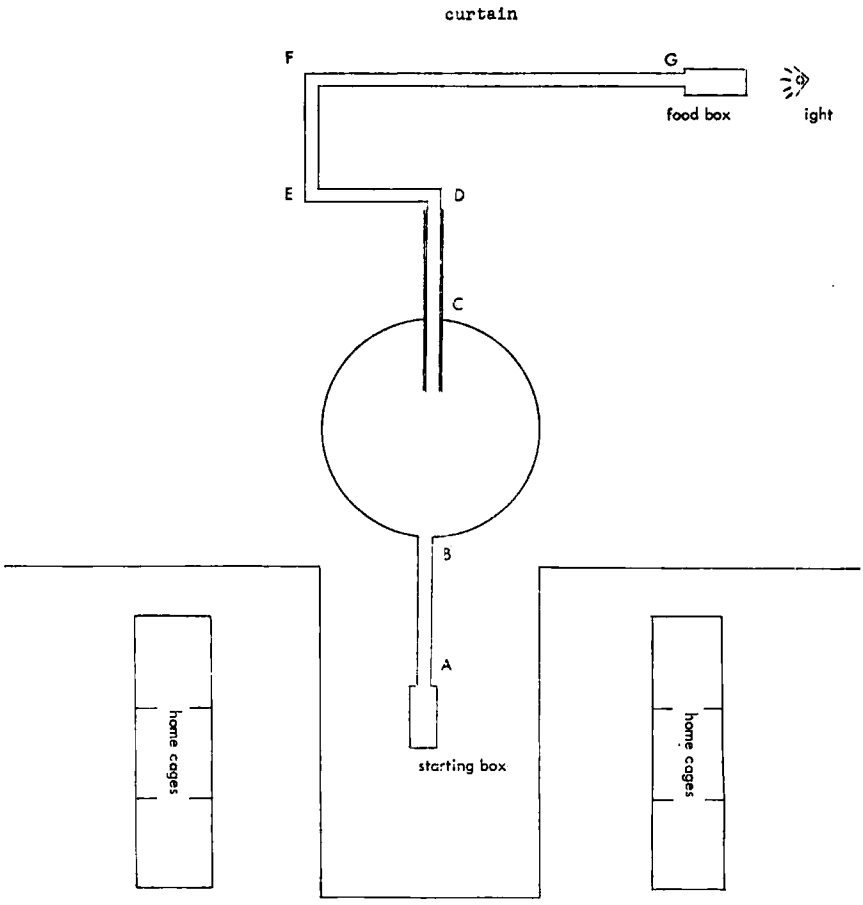


FIGURE 1
TRAINING APPARATUS

box from the middle of *FG*, and on the third trial each animal was started from *F*.

b. Day 2. On the first trial they were again run from *F*. On the second and third trials, the animals were put into the entrance of, and prodded through the alleyway to the food-box.

c. Day 3. On the first trial the animals were again forced to run through *CD*, but on the second and third trials they were placed on the starting pathway and were permitted to proceed independently to the table and through the alleyway to the food-box.

d. Day 4. The animals were given three more practice trials starting from *A*.

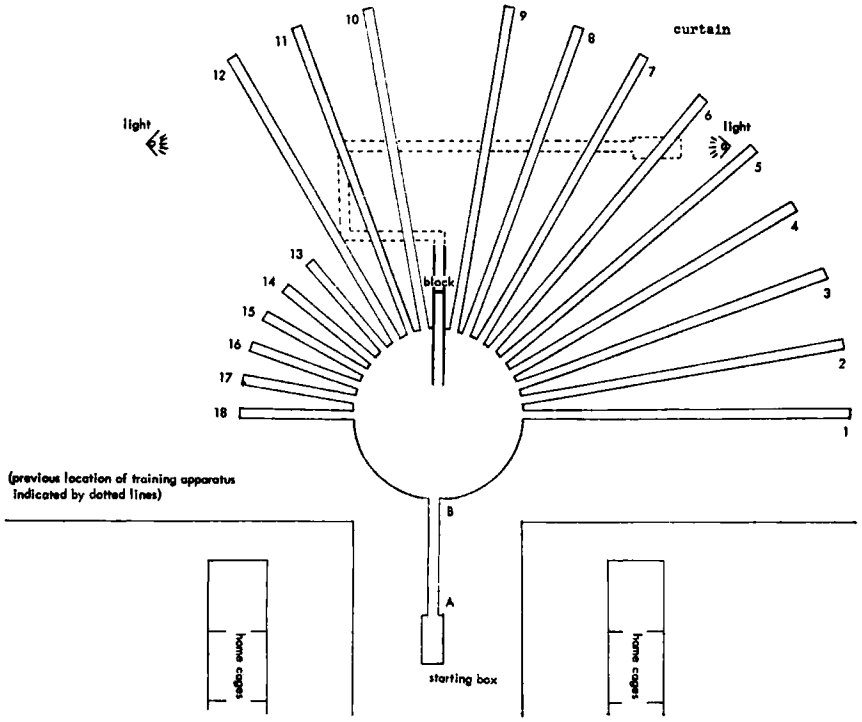


FIGURE 2
TEST APPARATUS

2. *Test*

On the fifth day the test situation was presented and the animals were randomly divided into three groups of 12.

Group 1 was tested in the same manner as described by Tolman, Ritchie, and Kalish (12), with the lamp, used in training, remaining in the same position.

Group 2 was tested in the same way with, however, a lamp correspondingly placed on the opposite side of the apparatus, in place of the originally used light.

Group 3 was required to run the test trial with both lamps lighted. After each test trial the table top was rotated 90 degrees. The food-box was removed before beginning the test trials.

3. *Results and Discussion*

Before dealing with the results of the first experiment, it is necessary to explain some of the terms used to describe and to categorize the animals' behavior patterns. The expression final path or choice has already been defined as the first pathway which an animal traverses completely. Since, in this experiment, an animal was removed after making a final choice, or after six minutes had elapsed without a choice having been made, a maximum of 12 final choices could be recorded for any one group. The initial choices referred to, signify the *first* pathway each animal entered. An entry was recorded when an animal ran out a full body length onto a pathway. In most cases, the initially chosen pathways were only partially traversed. In a few instances, however, animals went to the end of the initial pathway, or subsequently returned to it and made it the final choice as well. The total number of entries is a record of *all* the pathway entrances the animals made, including the initial and final choices.

TABLE 1*

	Group I			Group II			Group III		
Response	Total no.	No. on paths— 5-6-7	12-13-14	Total no.	No. on paths— 5-6-7	12-13-14	Total no.	No. on paths— 5-6-7	12-13-14
Entries	48	2	5	46	5	3	34	4	2
Initial	11	2	3	11	1	1	11	0	2
Final	7	0	1	5	2	1	5	1	0

*A comparison of choices made on Paths 5-6-7 and 12-13-14 after four days of training.

The results of the experiment are summarized in Table 1. Upon initial examination, these results reveal that the behavior of the animals in none of the three conditions used for testing is in accord with the behavior that would be predicted by a cognitive theory. Nor do the results provide immediate evidence which is favorable for the sensory-integration viewpoint. The animals in the test situation do not respond either to a goal-place orientation, or simply to a light stimulus. The animals in Group 1, tested with the position of the lamp remaining unchanged, showed no greater disposition to traverse Paths 5, 6, and 7, the goal-place directed paths than to enter Paths 12, 13, and 14, corresponding paths on the opposite side of the appa-

ratus. The animals tested with the lamp on the left or opposite side, or with both lamps lighted, Groups 2 and 3, respectively, showed the same lack of preference for Paths 5, 6, and 7 leading to the original food place, and for the light directed down the corresponding Paths 12, 13, and 14 on the opposite side of the laboratory from the original food place. It is evident that relatively few choices were made on either of the path groups enumerated. Thus, in none of the test groups, nor in any of the response categories is there as yet any real indication of either specific food-place orientation, or of light directed behavior.

The preponderance of responses, indicating a general persistence in behavior, occurs on Paths 8 and 9, or on Paths 10 and 11. These four pathways, which make up only 22 per cent of the available choices, encompass 55 per cent, 52 per cent, and 51 per cent of the total number of entries made by each of the three test groups. The number of entries on Paths 5, 6, and 7 together with those made on 12, 13, and 14, never exceeded 18 per cent of the total entries, a percentage which is less than the theoretical expectancy for random choices. It appears then that the animals select the center pathways, which are closest to the training path, with sufficient frequency to be in apparent accord with a response learning theory.

However, it would be premature to conclude that a response learning theory is supported by the data, because thus far we have considered only the *end results* of the animal's behaviors. To understand what processes are involved in the learning it is necessary to analyze the behaviors of the animals in detail and to consider the complexities of the problem situation. Such an examination indicates that a response learning theory is only apparently adequate and can be retained only so long as consideration of the data is operationally restricted to the end-results of the animal's behaviors. A more complete analysis does not support response theory. An analysis of the structure of the problem reveals certain complexities that contribute to the "response-like" behavior of the animals.

In the original learning situation, the animals ran from the starting pathway, onto the table, and into Alley *CD* which was completely darkened by the high alley walls. After running through the canyon-like pathway, the animals had to make a sharp turn to the left, away from the food and light. That is, they were required to run toward the darkened side of the room. After this turn was made, the animals proceeded to turn in the direction of the light, and toward the food place. The final pathway required the animal to make a right-angle turn to the goal. In the early stages of the training, many animals gave evidence of experiencing difficulty in turning

away from the light and goal after leaving Alley *CD*. They manifested much more hesitant running, and even long delays, in running at this point. The animals frequently reared up in the direction of the goal-box before turning to the left, away from the food place. Bearing these factors in mind, the data on the preference for the center pathways must be subjected to closer scrutiny.

The data presented in Table 2 point to a dichotomy between Paths 8-9 and 10-11. This breakdown would also seem justified by the situation itself, for Paths 8 and 9 lead toward the side where the food and light were located in the training situation, and Paths 10 and 11 lead in the opposite direction, or in the direction of the turn following Pathway *CD*. Also Paths 10-11 were shadowed by the high walls of *CD* when the light was unchanged and Paths 8-9 were brightly illuminated.

TABLE 2*

	Group I			Group II			Group III		
Response	Total no.	No. on paths— 8-9	10-11	Total no.	No. on paths— 8-9	10-11	Total no.	No. on paths— 8-9	10-11
Entries	48	5	18	46	13	11	34	6	11
Initial	11	1	4	11	3	3	11	1	5
Final	7	0	4	5	1	0	5	0	4

*A comparison of choices made on Paths 8-9 and 10-11 after four days of training.

The data of Table 2 further indicate the trends which seem to be underlying the behavior of the animals in the test situation. In the group that was tested with the position of the light remaining unchanged, Group I, five of the 48 entries made were on Paths 8-9, the lighted paths, while 18 of the entries were made on Paths 10-11, the darkened paths. This same relative preponderance of responses is found in the data on the initial entries—only one of the 11 initial choices made by this group was on the former pair of pathways, and four were made on Paths 10-11. The final choices made by this group also show a preference for these two paths, for four of the seven final choices made were on Paths 10-11, and none were made on Paths 8-9.

The data on the second group, animals that were tested with the lighted lamp on the left, or on the opposite side of the laboratory from the original food place, show a generally equal distribution of responses between the two groups of pathways. In this group, 13 and 11 of the 46 entries made were on Paths 8-9 and 10-11, respectively. Of the 11 initial choices made, these pairs of paths were each selected three times. Paths 8 and 9 encom-

pass only one of the five final choices made, and Paths 10 and 11 were not selected at all. The group tested with both lamps lighted again shows the general preference for Paths 10 and 11 with, however, a decrease in the activity level of the animals. This group, as a whole, made fewer entries than either of the other groups, but the data does indicate a tendency to run on Paths 10 and 11. Of the total of 34 entries made by this group, six were on Paths 8-9, and 11 were on Paths 10-11. The former pair of pathways comprised only one of the 11 initial choices made, as compared with five on Paths 10-11. The final choices show this same marked preference for Paths 10 and 11, as four of the five final choices were made on this group of pathways, and none were made on Paths 8 and 9.

That the learning resulting from the training situation was not complete is fairly well evidenced by the very few instances of rats responding to the food place or to the light stimulus. Thus the animals had not quite mastered the problem in responding directly to these goal factors in the maze situation. That they had learned something is well indicated by their behavior during the learning, or training situation, by the rapidity with which they traversed the training maze, and by their speed in initiating the performance pattern by entering the alley *CD* in the test trials. What, however, had the animals learned, and to what were they responding in the test situation? Since the light stimulus was the only factor varied in the three experimental situations, an explanation of the behavioral differences elicited must be based on this—either alone or in combination with other factors. If the animals were responding only to the light, then there should have been no definite differences between the behavior of Groups I and II in the test situation—each should have oriented directly to the light, with the first group going to the right and the second group to the left. The third group should have made either more random choices, or should have made *no responses* at all. On the other hand, if the underlying orientation was based mainly on kinesthetic cues, then all three groups should have responded in the same way—with the directional trend being the most dominant factor, regardless of the light conditions. Neither of these conjectures, however, bears any significant fruit. The resolution and explanation of the characteristic responses of the three groups, therefore, must be based upon a number of factors. It would seem that the animals had made a twofold sensory investment in the learning of the original maze problem. The learning had involved, primarily, a turn to the *left* and away from the light. Group III, the animals tested with both lamps lighted, a visual condition in which the light stimulus is negated as a response cue, predominantly selected the left of center path-

ways, Paths 10 and 11, and there was a considerable drop in the total number of entries made. When Group I was tested, with the position of the light remaining unchanged at the right, the animals were able to respond towards the left, and at the same time, away from the light—resulting in a preponderance of responses on Paths 10 and 11. In the second group, however, these very favorable conditions were changed in such a way that the animals were faced with a conflict. Since in the test situation the light was now placed on the left side of the maze, they could only respond *either* toward the left *or* away from the light. Therefore, the two sensory cues have opposed characteristics as direction determines. The prediction one would make on the basis of this hypothesis is borne out and the data present an even distribution of responses around the center path choices. Paths 8-9 and 10-11 are selected with almost equal frequency by the animals in this group. Thus, we find here a preponderance of responses on all four center pathways with an almost equal grouping on each pair of pathways, a further indication that in this instance, some animals were responding with the left turn tendency, and others were responding away from the light visually. The behavior of the animals in the third group, tested with both lamps lighted, bears out these contentions still further. Here the visual factor is negated as a cue, for no turn can be made away from the light, and the animals, so it would seem from the data, are forced, or return, to the kinesthetic or directional trend—and the left turn again becomes dominant.

The data of the first experiment indicate that two kinds of sensory integration may be developing. On the one hand, the animals appear to have developed a kinesthetic organization which is derived from the structure of the training maze, while, on the other hand, they are beginning to respond to light as equivalent to the food. It was felt, therefore, that extending the training would permit changes in the sensory organization to develop which would throw light on the cues which dominate in determining the animal's behavior and would help to reveal the factors underlying the choices that are made in the test situation. Consequently, 10 additional days of training were given and the evidence presented below derives from tests which were administered after these additional days of training.

F. Experiment II

Sixty-two days after Experiment I was completed, an extension of the previous program was instituted, employing the same 36 animals. The apparatus used in both the training and test situations was identical with that used in the first experiment.

1. *Training*

The animals were given 10 days of additional training. On the first day each was given three trials, the first from *F* to the goal and two more from *CD*. On the second day the animals were given two more trials starting from *CD*, and another from *A*, the starting pathway. On the next five days the animals were given two trials per day starting from the initial pathway. On the final three days they were given three trials per day from *A*. Thus, the animals had made 20 additional complete traversals of the training maze.

2. *Test*

On the eleventh day, the test day, the sun-burst maze, Figure 2, was substituted in the same manner, as in the first experiment and with the same test-group arrangements as had been previously used. The populations of the respective groups were identical in both this and in the previous test situation. Each animal, therefore, was tested under the same experimental conditions it had experienced earlier.

In both Experiments I and II a record was made of the time expended by each of the animals in traversing the apparatus during the various training trials. On the test days an additional record was made of the paths each of the animals entered as well as the pathway, if any, it completely traversed. As soon as a complete traversal was made, or after six minutes had expired, the animal was returned to its home cage.

3. *Results of Experiment II*

When the final choices of the three groups of animals are compared (cf., Figure 4) it is found that the light variable has now become more influential in determining the performances of the animals in each of the three experimental conditions. In Group 1, the animals were tested with the position of the lighted lamp remaining unchanged. In this group six of the eight animals that made final choices selected the right, or lighted side of the sunburst, which, in this instance, coincided with the original food side. It is significant to note that the two animals in this group that responded differently from the other animals spent more time scurrying about on the test maze than the other animals that made final choices, before making their choice toward the left in accordance with kinesthetic cues—the characteristic response found in Experiment I. When, however, the light stimulus was moved to the left, to the side opposite the original food place, nine, or all of the animals in Group 2 that made final choices, selected pathways on the left or lighted side of the sun-burst maze, despite the fact that this

was in the direction away from the place at which they had formerly obtained food. The final choices of the group of animals tested with both lamps lighted, Group 3, were evenly split between the two sides of the apparatus, with these animals making the fewest final choices of all three groups in this experiment. Three of the five final pathways chosen were on the left side and two were on the right side of the apparatus. This low num-

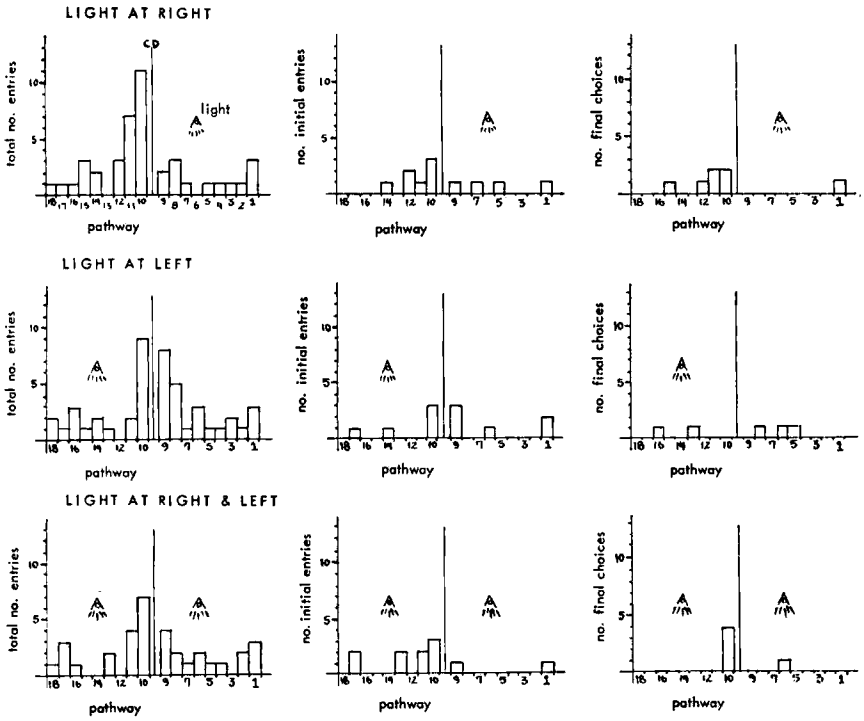


FIGURE 3
PERFORMANCE AFTER FOUR DAYS OF TRAINING: EXPERIMENT I

ber of final choices when taken in conjunction with the fact that this group of animals made the greatest number of entries indicates that conflict did arise. Also, fewer animals were able to resolve the situation by going toward either of the lights. This is demonstrated by total and initial entries as well as by final choices. It should be noted that the direction of the original food place added no significant weight in determining either entries, initial choices, or final choices.

Consideration of these final choices reveals that the additional training

has caused a fundamental modification in the behavior as compared with the results of the first experiment. In the first experiment, the kinesthetic orientation was the dominant one and the animals responded, in the main towards the left, with a reinforcing tendency to turn away from the light stimulus. Thus, when the lighted lamp in the test trial remained in its original position at the right, or when the two lamps were employed, a situa-

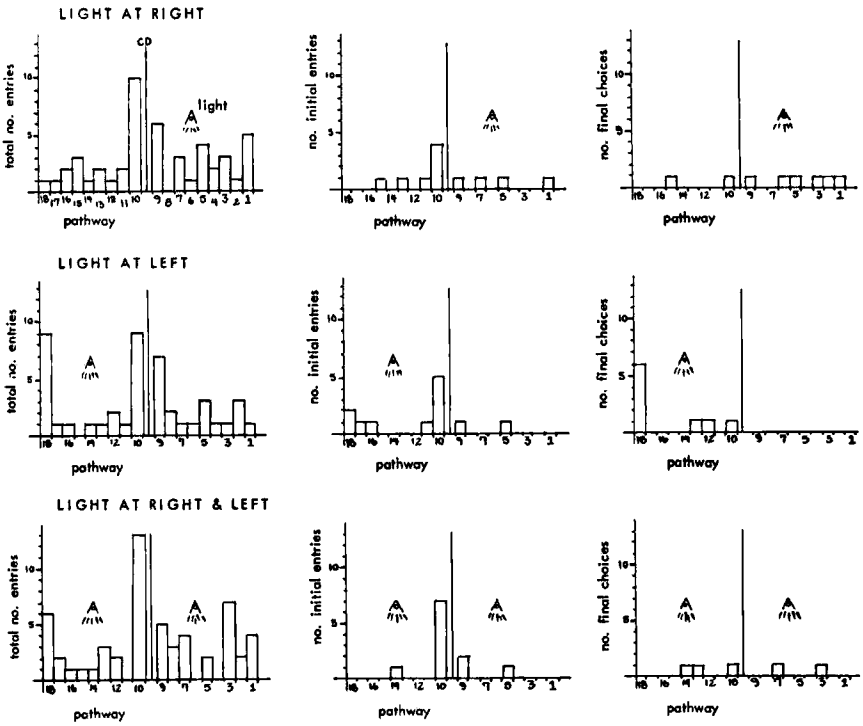


FIGURE 4
PERFORMANCE AFTER 10 ADDITIONAL DAYS OF TRAINING: EXPERIMENT II

tion which vitiated light as a discrimination cue, the animals responded, in the main toward the left side of the laboratory—towards the side away from the original food place. When, however, the light stimulus was moved to the left side of the apparatus, as was the case for the animals in Group 2, the responses were evenly distributed with some animals responding to the left, and some responding with a right turn, leading away from the light. This bi-modal tendency supports the idea that at the very least a twofold sensory investment is being made by the animals in the learning situation.

In the second experiment, however, the extended training preceding the test series yields a heightened effectiveness of the limit cue, and the kinesthetic cue of the left turn, *per se*, is no longer the dominant orientation underlying the animals' choices on the sun-burst maze. With the prolonged training period the animals in each of the two groups tested with a single light responded toward their respective light cue even when this light, as for Group II, was on the left side, the side opposite to its original place in the training situation. The third group, on the other hand, tested with both lamps lighted, responded towards either of the lights, with no evidence of a selective orientation towards the original food place—half of the animals went to the left and half to the right.

Although the continued attractiveness for Path 10 in the data on the entries and initial choices of all three groups seems to hark back to the trends suggested by the first experiment (the comparative data may be seen in Figure 4), this contention is not borne out strongly in the other response category. The tendency of the animals to avoid the four center-pathways—they now comprise only four of the 22 final choices whereas after the first experiment they encompassed nine of the 17 final choices made by all three groups of animals—indicates that the animals in Experiment II are probably responding at a different level from that which was previously encountered in Experiment I. The data on final choices which has already been discussed in connection with the position of lighted lamps in the test situation for the different groups of animals also makes this quite evident.

Moreover, a comparison of the data gathered in the two experiments with regard to the selection of Paths 1 and 18 highlights the behavior differences that result from the extended training period. In the first experiment, Path 18 was responded to very infrequently and it encompassed not more than 4 per cent of the total number of entries of any of the three experimental groups and was not chosen as an initial or final path by any of the animals (cf. Figure 3). Path 1, on the other hand, comprised between 7 and 9 per cent of the total number of entries made by the various groups and was chosen as an initial pathway only once by animals in Groups 1 and 3 and twice by animals in Group 2. Only one animal, in Group I, selected Pathway 1 as a final choice. In the second experiment an indicative change occurs. The data on the responses of Groups 1 and 3 remain relatively consistent with only slight changes occurring in the latter group (cf. Figure 4). The significant change, however, occurs, in the responses of Group 2, tested with the lighted lamp on the left side of the apparatus. In this group, Path 1, leading toward the side of the original food place and away from

the light in the test trial, is chosen in only 2 per cent of the total number of entries (one out of 45 entries) and is not selected as an initial or final choice. Path 18, on the other hand, leading at a right angle from the starting pathway toward the *lighted* side of the laboratory which in Experiment I registered only 4 per cent of the entries (two out of 46) now makes up 20 per cent (nine out of 45) of the total number of entries and encompasses 17 per cent of the initial (two out of 12) and 67 per cent of the final path choices (six out of nine) made by this group of animals.

G. Some General Problems of Theory

Considered in a broad manner the experiments reported in this paper have a twofold significance. On the one hand they reveal, that in the "sun-burst" situation which was specifically designed to study "place" learning and the formation of goal "expectancies," the behaviors that are obtained can be accounted for in terms of the formation of stimulus equivalences and not by either a "cognitive" or by a "response" theory of learning. On the other hand, the study suggests that "cognitive" learning theory and "response" learning theory are neither truly antagonistic viewpoints nor *methodological* alternatives for explaining behavior. It is this general consideration that demands some concern, because if the two viewpoints are not real alternatives, then a whole body of experimental evidence that has arisen as a result of the *supposition* that "response" and "cognition" represent opposing explanatory devices must be reassessed.

The results of the experiments we have reported indicate that neither a response theory of learning nor a cognitive map-place theory is adequate to explain the behavior of our animals. A longitudinal view of the behaviors shown in the sun-burst situation shows that the animals were *responding to different aspects of the situation at different points or stages in their learning.* At least two sets of sensory cues contributed to the elaboration of the learned pattern. One of these was the kinesthetic complex produced by the necessity for making turns. The other cues to which the animals were responding were visual in nature, and were derived from the differential brightness pattern produced by the light at the goal-box and the shadowing effect of the high entrance-alley wall. An understanding of the animal's behaviors can only be obtained if one analyzes the manner in which continuing experience with the situation causes the animal to be dominated by or to utilize different aspects of the stimulus complex in making its responses. Once such an analysis is made (as in the preceding section of this paper) it becomes clear that neither cognitive theory nor response theory is dealing

with the learning process, but that each represents an *ad hoc* explanation of the behavior an animal may exhibit at a given stage or level of its learning. The "place" concept deals *only* with the performances shown when the animal is predominantly responsive to light cues, while the "response" conception concerns itself primarily with the behaviors that are aroused by the kinesthetic cue structure of the situation.

Therefore the two theories do not represent truly opposing explanations of the same behaviors. On the contrary, the behavior one view is trying to explain is quite separate from the pattern of activity that has engaged the attention of the other. Consequently, there have arisen two separate bodies of "evidence" each independent of the other. As a result representatives of each school have by: (*a*) excluding different "negative" instances or (*b*) terminating an investigation at different "favorable" points, or, (*c*) constructing a situation in which one or another cue system is favored, or, (*d*) using animals having different life histories and consequently different patterns of responsiveness, readily obtained experimental evidence which appeared to favor a given theoretical viewpoint.

As a consequence of these tendencies, a new kind of parsimony is developing in learning theory. It is not the parsimony of unwarranted assumptions enjoined by Occam and Morgan. On the contrary, it represents the bold elaboration of theory through the parsimonious accumulation of facts. By sufficiently limiting a situation and by implicitly assuming it to be *typical* of *all* the circumstances in which learning takes place, "theories" of learning have been developed which can best be described as situational. Data obtained under other conditions is referred to only rarely and paucity has come to replace parsimony as a guiding principle.

Tolman (11), in his consideration of whether there is more than one kind of learning, has shown an awareness of the possibility that different learning "theories" may have been derived from the observation of behaviors in situations which make different demands on the organism. From this, he has concluded that probably every theory has some merit. However, it is possible to arrive at a different conclusion, namely, that none of the theories has sufficient merit and generality to warrant being called a theory of learning. The elaboration of this proposition will be the burden of a subsequent paper.

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